Quadratic equations

1 The equation $x^2 - 144 = 0$ has a positive integer solution of $x = 12$. Find its other solution.

2 Solve the following equations:

a $x^2 = 2$ **b** $x^2 = 25$ **c** $x^2 = 121$ **d** $x^2 = 294$

e $x^2 - 121 = 0$ **f** $x^2 - 10 = 15$ **g** $\frac{x^2}{16} - 2 = 2$ **h** $\frac{x^2}{25} - 3 = 6$

i $(x+3)^2 = 49$ **j** $(x-3)^2 = 64$ **k** $(x-6)^2 = 2$ **l** $(2-x)^2 = 81$

m $(x-7)^2 = 81$ **n** $(7-x)^2 = 81$ **o** $(8x+9)^2 = 256$ **p** $81x^2 - 16 = 0$

3 Solve the following equations:

a $x(x+7) = 0$ **b** $x(2x-9) = 0$

c $(10x-9)^2 = 0$ **d** $(4x-9)^2 = 0$

e $(-3+7x)^2 = 0$ **f** $(x-4)(x-2) = 0$

g $(x-6)(x+7) = 0$ **h** $(8x-5)(3x-7) = 0$

i $(3x+8)(5x-7) = 0$ **j** $(3x-17)(2x-d) = 0$

4 Solve the following equations:

a $4y^2 = 100$ **b** $25y^2 = 36$

c $-3k^2 = -12$ **d** $81k^2 + 8 = 24$

e $-25v^2 + 64 = 0$ **f** $10\left(p^2 - 7\right) = 930$

g $4m(m+5) = 0$ **h** $\frac{m}{2}(m+5) = 0$

5 Solve the following equation for x, in terms of a and c. Assume a and c are positive.

$$ax^2 - c = 0$$

Applications

6 The equation $4x^2 + kx + 16 = 0$ has one solution: $x = 2$. Find the value of the coefficient k.

7 The equation $ax^2 - 32x - 80 = 0$ has one soulution: $x = 4$. Find the value of the coefficient of a.

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8 The Widget and Trinket Emporium has released the forecast of its revenue over then next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation:

$$R = -(t-12)^2 + 4$$

When will the revenue be zero?

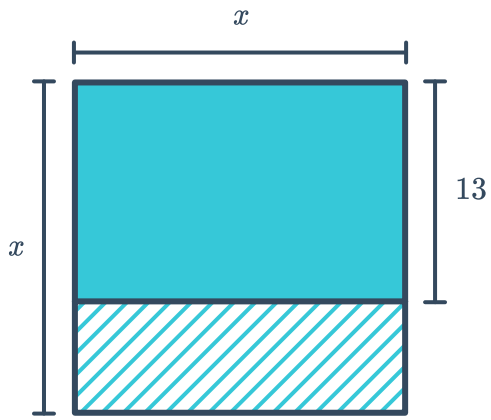
9 Neville needs a sheet of paper $x\,\text{cm}$ by $13\,\text{cm}$ for an origami giraffe. The local origami supply store only sells square sheets of paper.

The lower portion of the image below shows the excess area A of paper that will remain after Neville cuts out the $x\,\text{cm}$ by $13\,\text{cm}$ piece. The excess area, in square centimeter, is given by the equation:

$$A = x(x-13)$$

a At what lengths x will the excess area be zero?

b For what value of x will Neville be able to make an origami giraffe with the least amount of excess paper?



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